温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Descriptive Statistics II

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Recap & Transition

On Tuesday, we learned how to describe a single variable:

- **Location** → mean, median, mode
- **Dispersion** → range, percentile, IQR, variance, standard deviation
- **Shape** → skewness and kurtosis
- **Outliners** → identification and treatment

Today, we move one step further:

- Instead of describing variables **individually**,
- We ask: *How are two variables related?*

➡️ This brings us to Measures of Association:

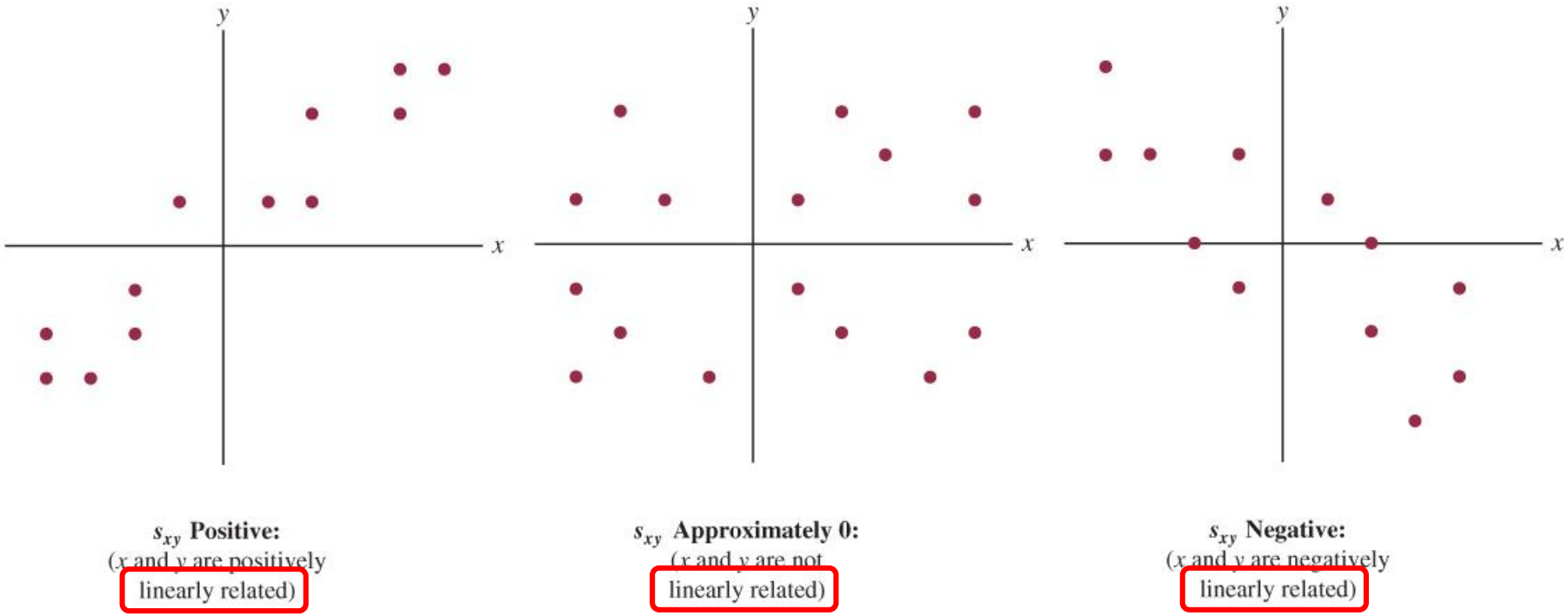
- **Covariance** → direction of relationship
- **Correlation Coefficient** → direction *and* strength of relationship

Measures of Association – Covariance

- Definition
  - A measure of the **liner association** between two variables, X and Y.
- Formulas
  - For a population $$cov(X,Y) = \frac{\sum_{i=1}^{N}(x_i - \mu_x)(y_i - \mu_y)}{N} = \sigma_{xy}$$
  - For a sample $$cov(X,Y) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n-1} = s_{xy}$$
- Intuition
  - Indicate whether two variables move together
    - Positive covariance: both increase together
    - Negative covariance: one increases, the other decreases
    - Zero covariance: **no linear relationship**
- Example
  - Height & weight → positive covariance
  - Price & demand → negative covariance

Measures of Association – Covariance

Scatter Charts and Associated Covariance Values for Different Variable Relationships



Source: Camm, Ch2

Measures of Association – Covariance

- Two Questions
  1) Can we calculate covariance between two variables with different lengths?
     - ❌ No — X and Y must have the **same** number of **paired** observations.
  2) Can we compare covariance values across different datasets?
     - ❌ Not directly — because covariance depends on the **scale of measurement**.
- Lesson
  - Units = product of X and Y units
    - Example: height (cm) × weight (kg) → covariance in $\mathbf{cm \cdot kg}$
  - Magnitude depends on measurement scale
    - Doubling all height values would double the covariance
- Limitation
  - Covariance indicates **direction** only (positive/negative)
  - Magnitude is not standardized → cannot compare across datasets
- Example
  - Cov(Height, Weight) = 200 (cm·kg)
  - Cov(Income, Education Years) = 15,000 (USD·years)
  - Which is “stronger”? We cannot tell, because the units differ.

Measures of Association – Correlation Coefficient

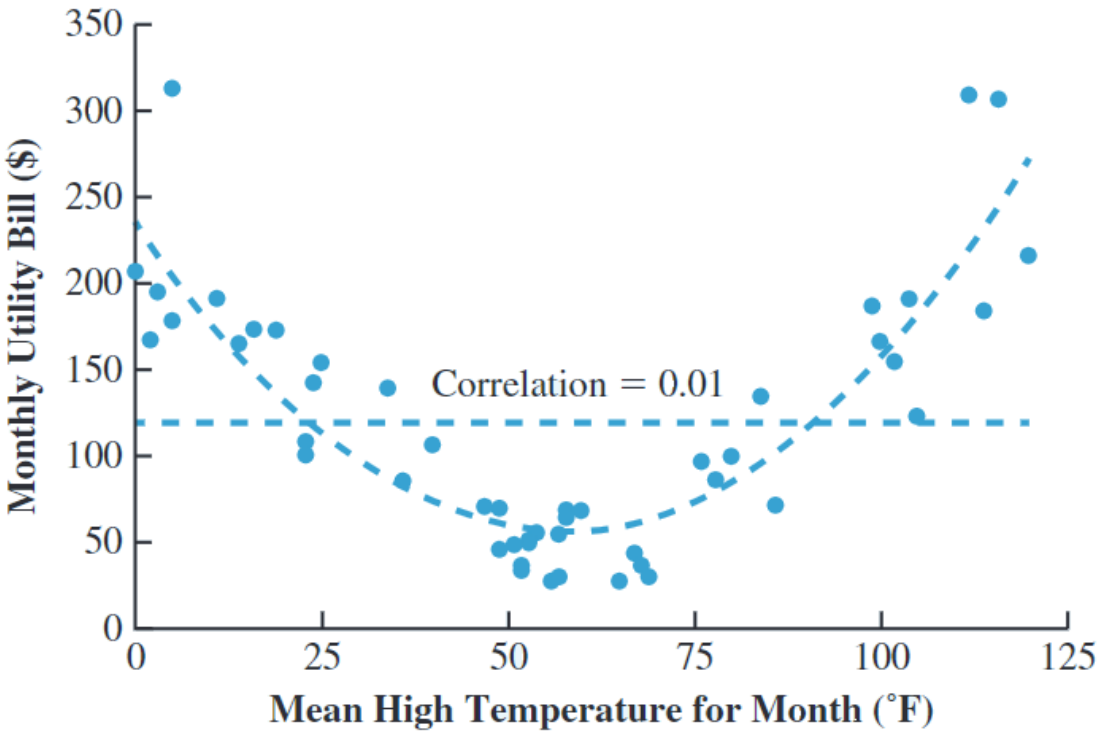
- Definition
  - The correlation coefficient measures the **strength** and **direction** of a linear relationship between two variables, X and Y.
  - Unlike covariance, it is **standardized** and **unit free**.
- Formulas
  - Population correlation (ρ)
  - Sample correlation (r)

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} \qquad r_{xy} = \frac{s_{xy}}{s_x s_y}$$

- Range & Interpretation
  - Range: $\boldsymbol{-1 \leq r \leq +1}$
  - Closer to +1 → stronger positive linear relationship
  - Closer to -1 → stronger negative linear relationship
  - Closer to 0 → weaker liner relationship
  - Also known as Person product-moment correlation or Pearson's correlation coefficient

Measures of Association – Correlation Coefficient

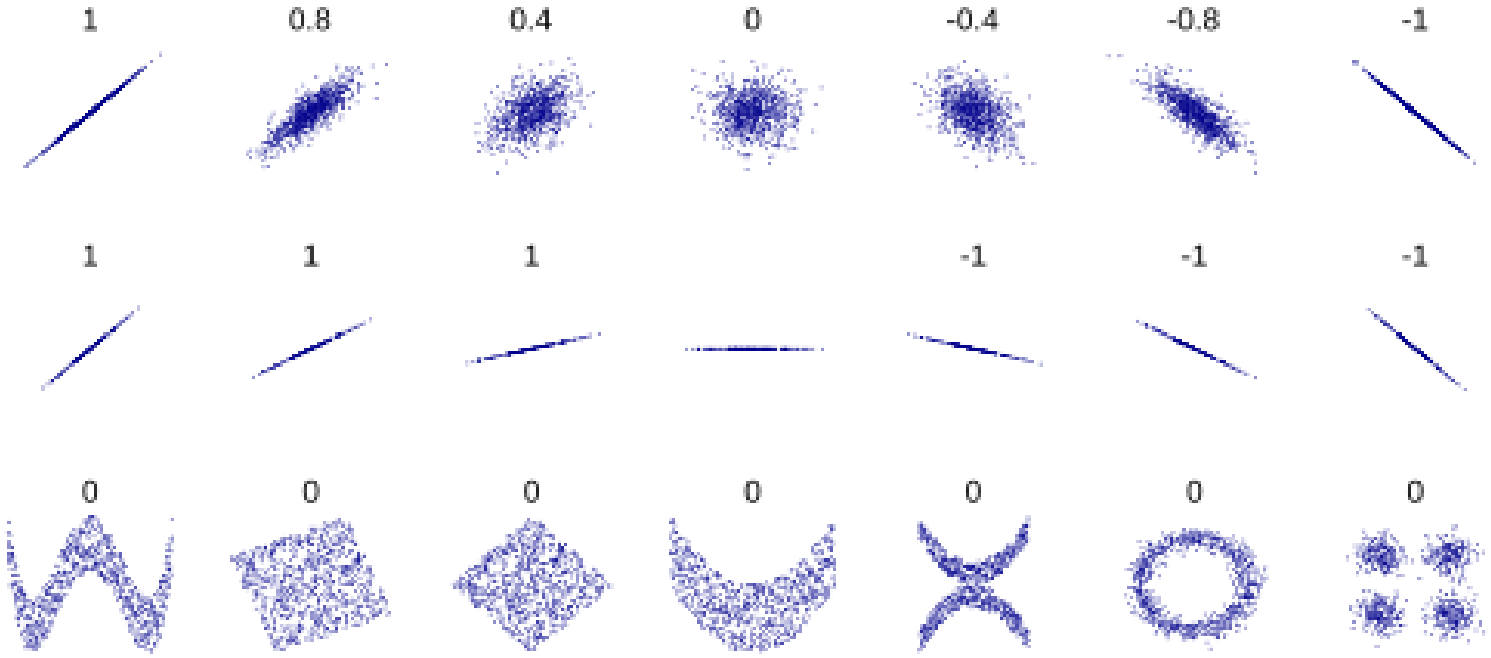
- Correlation coefficient measures only the *strength* and *direction* of the **linear association** between two quantitative variables.
- It does not capture **nonlinear** relationships.
- A correlation near 0 means no linear association, but variables could still be strongly related in a nonlinear way.
- Example: $r_{xy} = 0.01$ indicates no linear relationship, yet the scatterplot shows a clear **nonlinear** relationship.
- Lesson
  - Always combine numerical measures (like r) with visual inspection (scatterplots).
  - Correlation ≠ causation, 0 correlation ≠ "no relationship" — it means no linear relationship.



Source: Camm, Ch2

Measures of Association – Correlation Coefficient

Several sets of (*x*, *y*) points, with the Pearson correlation coefficient of *x* and *y* for each set.



Source: Wikipedia (Correlation)

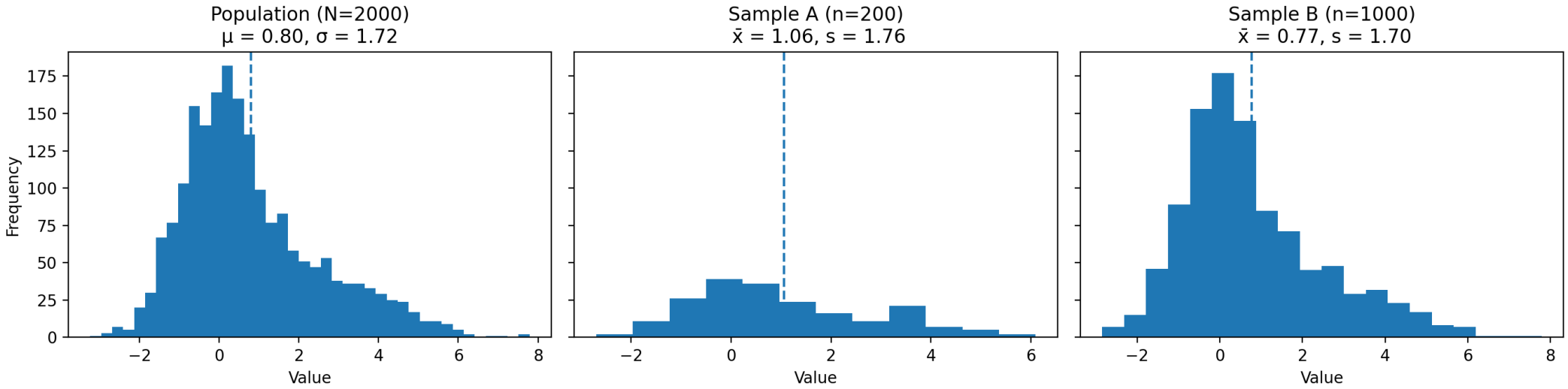
- The correlation reflects the **noisiness/strength** and **direction** of a linear relationship (top row).
- It does not capture the **slope** of that relationship (middle), nor many aspects of **nonlinear** relationships (bottom).
- The figure in the center has a slope of 0 but in that case, the correlation coefficient is undefined because the *variance of Y is **zero***.

Statistical Thinking in Business Decision-Making

- Statistical Thinking is a philosophy of learning and action for improvement, based on principles that:
  - all work occurs in a system of interconnected processes
  - variation exists in all processes
  - better performance results from **understanding and reducing variation**
- Business Analytics provide managers with insights into facts and relationships that enables them to make better decisions.

Variability in Samples

- Different samples from any population will vary.
  - different means, standard deviations, and other statistical measures
  - differences in shapes of histograms
- **Samples are extremely sensitive to the sample size** – the number of observations included in the samples.



Mini Task: Descriptive Statistics with Python

- Dataset: “dataset_*students_scores.csv*” from Canvas
- What to do
  - One variable

    mean, median, mode, min/max, range, var, std, quantile(Q1/Q3), IQR, skew, kurt
  - Whole DataFrame at once

    .describe(), .var(), .std(), .cov(), .corr()
- Functions to remember

  Series.mean/median/mode/var/std/quantile/skew/kurt

  DataFrame.describe/var/std/cov/corr

Mini Task: Outliner Detection Practice

- Purpose
  - Learn how to detect and handle unusual values in a dataset.
  - Apply both numeric rules and visual inspection.
- Methods Covered
  - Z-score rule → flag values with $|z| > 3$.
  - IQR rule → flag values outside $[Q1 – 1.5\times IQR, Q3 + 1.5\times IQR]$.
  - Visualization → histograms and boxplots to spot outliers.
- Dataset
  - Includes an unusual low MathScore and an extreme Height value.
  - You will judge whether these are true outliers.
- Your Task
  1. Apply Z-score and IQR rules to detect outliers.
  2. Use histograms and boxplots to confirm visually.
  3. Decide: Keep, remove, or correct? Justify your choice.